

QVLM

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≡ Idea (1 line)	NeurIPS'24
≡ Tags	PTQ

Background

- Large vision-language models (LVLMs)
 - visual question answering
 - robot navigation
- Memory Overhead & Computation Cost $\Rightarrow$ Difficulty in deployment

Post-Training Quantization

기존 문제

기존 PTQ 방식: **Layer-wise Rounding Function Optimization**에서 두 가지 문제 존재

1. **NP-Hard 문제** [$\Rightarrow 2^L$]
2. **Cross-layer dependency 고려 X** [$\Rightarrow 2 \times L$]

The optimization objective J can be formulated as follows:

$$\begin{aligned} \min_{\{Q_k\}} \quad & J = \left\| W_q^{(n)} X_q^{(n)} - W_r^{(n)} X_r^{(n)} \right\|_2^2 \\ \text{s.t.} \quad & X_q^{(k+1)} = Q_k(W_q^{(k)} X_q^{(k)}) \end{aligned} \quad (1)$$

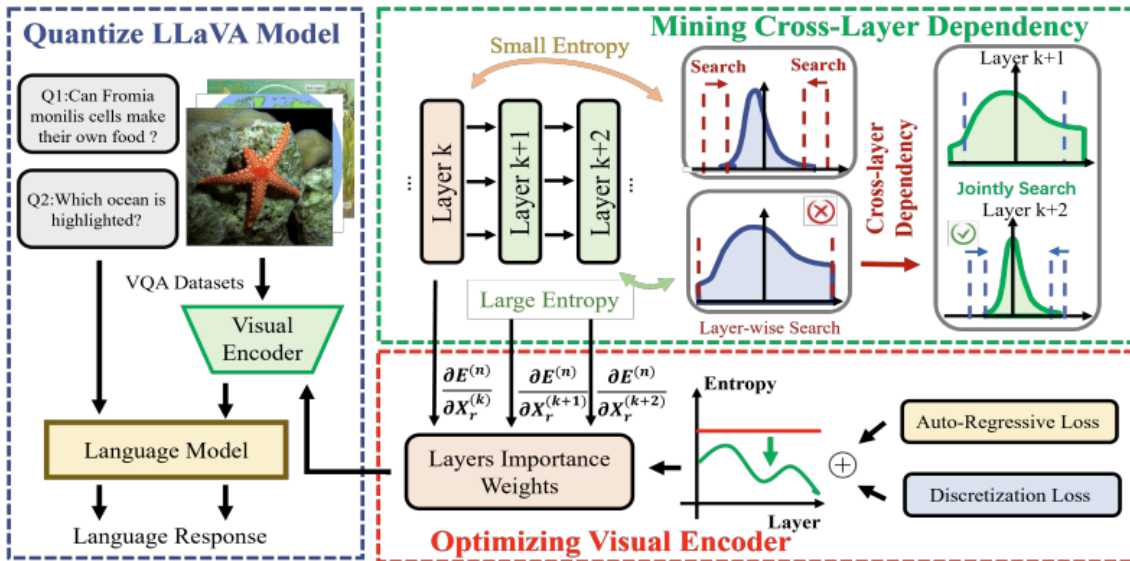
where $W_q^{(k)}$ and $X_q^{(k)}$ mean the quantized weights and activations for the k_{th} layer, and $W_r^{(k)}$ and $X_r^{(k)}$ represent their full-precision counterparts. Q_k means the rounding function for the k_{th} layer and n is the total number of the layers in the LVLM. Directly searching the optimal rounding function is NP-hard because the search space increases exponentially with the layer number. Therefore, conventional PTQ methods sequentially search the rounding functions by minimizing the quantization errors for each layer in the greedy way:

$$\min_{Q_k} \quad J = \left\| W_q^{(k)} X_q^{(k)} - W_r^{(k)} X_r^{(k)} \right\|_2^2 \quad (2)$$

where the layer index gradually increases to search the rounding function from bottom to top layers. However, the greedy search ignores the cross-layer dependency of discretization errors, which leads to accumulated discretization errors of model output even for the rounding function with small errors in the bottom layers.

$\Rightarrow$ 따라서, 시간(NP-Hard)와 Cross-layer dependency의 trade-off를 모두 고려한 **Block-wise Reconstruction** 제안

Overall Architecture



- Mining Cross-layer Dependency
- Visual Encoder Optimization

Quantization 순서

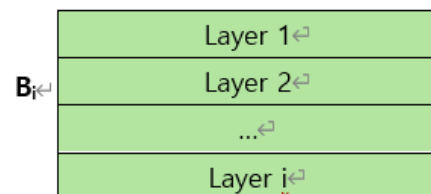
1. Layer 간 Dependency 계산
2. 1.에서 구한 Dependency를 활용한 Loss를 바탕으로 Visual Encoder Finetuning
3. Dependency 재계산 및 Block partition 구성
4. Joint Rounding Search를 통해 block 별 재구성

Mining Cross-layer Dependency

Block Recon 으로 VLM quant 진행하자

$$\min_{\{Q_k\} \in B_i} J = \left\| W_q^{(L_i)} X_q^{(L_i)} - W_r^{(L_i)} X_r^{(L_i)} \right\|_2^2$$

$$s.t. \quad X_q^{(k+1)} = Q_k(W_q^{(k)} X_q^{(k)})$$



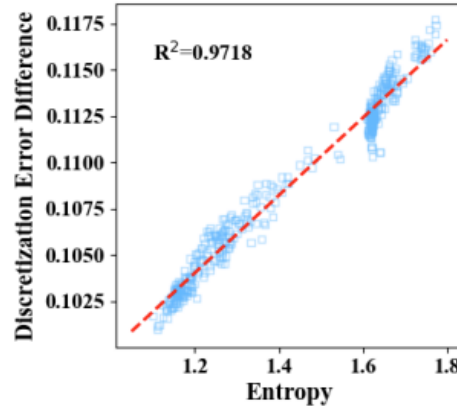
기존에는 각 layer 별로 했다면, **Block Recon**은 위와 같이 B_i 라는 block의 최종 output을 기준으로 quantization error 계산

그럼 어떻게 block을 나누는가?

- Block은 서로 강한 dependency를 가져야 함
- Cross-layer dependency를 직접 계산하면 LVLM을 여러 번 forward해야 해서 비용 ↑

따라서 dependency 대신 **proxy로 activation의 Information Entropy**를 활용

그럼, Proxy가 실제로 Dependency를 대변하는가?



x축: **Information Entropy (Proxy)**

y축: **Discretization Error Difference** $\Rightarrow$ layer-wise search와 joint search 간의 차이값

DED가 크다는 것은 joint search(block-wise)로 고려하는 것이 효과적 $\Rightarrow$ Dependency $\uparrow$
따라서 상관성 존재 (Proxy가 Dependency를 대변)

$$D(k, k+1) = - \sum_{ij} p(x_{q,ij}^{(k)}, x_{q,ij}^{(k+1)}) \log p(x_{q,ij}^{(k+1)} | x_{q,ij}^{(k)}) \quad (4)$$

$$p(x_{q,ij}^{(k)}) = \frac{\exp(-(x_{r,ij}^{(k)} - q_m)^2 / \Delta)}{\sum_{m=1}^M \exp(-(x_{r,ij}^{(k)} - q_m)^2 / \Delta)} \cdot \delta(x_{q,ij}^{(k)} - q_m) \quad (5)$$

$\swarrow$ k th layer의 (i, j) 가 해당 양자화 값을 가질 확률

$D(k, k+1)$: k 와 $k+1$ 번째 layer의 dependency proxy $\Rightarrow$ Cross-entropy

실제로는 RTN으로 인해 deterministic하게 반올림 양자화되므로 $p=10$ 이지만, Cross-entropy 안의 확률 변수 p 를 구하기 위해 Eq.(5)와 같이 확률 변수 추가

$$D(k_r, k_s) = - \sum_{k=k_r}^{k_s} \sum_{ij} p(x_{q,ij}^{(k)}, x_{q,ij}^{(k+1)}) \log(x_{q,ij}^{(k+1)} | x_{q,ij}^{(k)}) \quad (6)$$

$$B_i = \{ \bigcup_{k=k_-}^{k_s} Q_k | D(k_r, k_s) > (k_s - k_r) h_0 \} \quad (7)$$

멀리 떨어진 Layer의 경우에는 summation으로 dependency 계산

ex) $D(1,6) = D(1,2) + D(2,3) + D(3,4) + D(4,5) + D(5,6)$

그리고 block 별 최대 layer 개수를 정하고, average cross-layer dependency가 threshold보다 커야 block 구성

앞에서도 순차적으로 block 구성 진행

Optimizing Visual Encoders

Visual Encoder가 LVLM의 activation의 분포를 수정

따라서, Visual Encoder를 Quantization-friendly한 분포를 뽑도록 Calibration Set만으로 Fine-tuning 하자!

↵

Global Sensitivity: 현재 layer의 양자화 오차가
최종 출력에 미치는 영향↵

$$L_{ent} = \sum_{k=1}^n \left\| \frac{\partial E^{(n)}}{\partial X_r^{(k)}} \right\| \cdot \sum_{ij} p(x_{q,ij}^{(k)}, x_{q,ij}^{(k+1)}) \log p(x_{q,ij}^{(k+1)} | x_{q,ij}^{(k)}) \quad (8)$$

$$L_{err} = \|X_q^v - X_r^v\| + \eta \|X_q^{(n)} - X_r^{(n)}\| \quad (9)$$

↵

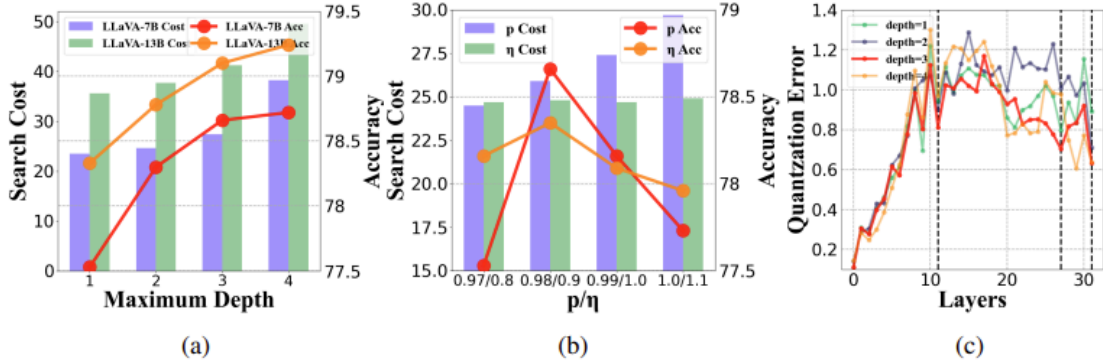
$$L = L_{reg} + \lambda_1 L_{ent} + \lambda_2 L_{err} \quad (10)$$

- L_{reg} : 기존 LVLM auto-regressive loss
- L_{ent} : Entropy loss, 전체 출력에 대한 민감도와 연속 layer의 local dependency를 고려하는 loss
- L_{err} : visual encoder와 LVLM 각각의 quantization loss

Experiments

Method	W6A6			W4A4		
	Memory	Search cost	Accuracy	Memory	Search cost	Accuracy
QLoRA	9.7G	23.7	88.43	6.6G	23.5	77.53
+CDM		26.9	88.95		25.9	78.66
+VEO		24.1	88.72		23.7	78.35
Q-VLM		25.1	89.34		24.6	79.79

Table 1: Effect of different LVLM quantization method we proposed. "CDM" means cross-layer dependency mining and "VEO" stands for visual encoder optimization. We report the result of LLaVA-7B model on ScienceQA dataset.



- Ablation: CDM + VEO
- Maximum Depth = 3



Which ocean is highlighted?

✗ INT4 AWQ
The Southern Ocean reaches from the shores of Antarctica to South latitude. **The answer is B.**

✓ INT4 Q-VLM
Oceans are huge bodies of salt water. The world has five oceans. **This is the Indian Ocean. The answer is A.**



Can Fromia monilis cells make their own food?
Options: (A) yes (B) no

✗ INT4 AWQ
Animals get their food by digesting other organisms. **But in the 1950s, scientists discovered that animals can make their own food.** Fromia monilis cells use chemosynthesis to make their food. **The answer is A.**

✓ INT4 Q-VLM
Today, many scientists classify organisms into six broad kingdoms. The table below shows some traits used to describe each kingdom. **Fromia monilis is an animal.** Animal cells cannot make their own food. **The answer is B.**

	Bits	Method	Subject			Context Modality			Average
			NAT	SOC	LAN	TXT	IMG	NO	
LLaVA-7B	FP	-	89.39	96.06	85.64	88.71	87.65	88.50	89.81
	W6A6	AWQ	85.39	92.01	83.27	84.80	83.54	85.99	86.23
		QLoRA	88.45	94.71	84.45	87.63	86.07	87.87	88.43
		Q-VLM	89.43	95.73	84.00	88.71	87.51	87.25	89.34
	W4A4	AWQ	74.33	72.22	74.82	73.41	67.13	77.98	74.02
		QLoRA	77.53	75.48	79.18	76.64	70.70	81.95	77.53
		Q-VLM	80.86	75.93	80.73	80.01	72.48	83.90	79.79
LLaVA-13B	FP	-	90.19	93.14	87.09	89.39	87.06	89.83	90.00
	W6A6	AWQ	88.03	92.60	84.00	86.02	85.18	86.41	87.57
		QLoRA	88.87	92.89	85.64	87.59	86.56	87.53	88.87
		Q-VLM	89.54	93.18	86.50	88.12	87.01	88.85	89.70
	W4A4	AWQ	80.71	70.61	78.49	79.46	70.76	81.82	77.91
		QLoRA	79.62	71.43	82.45	78.25	68.42	85.30	78.64
		Q-VLM	82.55	73.32	83.18	81.03	70.82	86.74	80.78
MoE-LLaVA-1.6B	FP	-	64.01	58.57	63.30	62.80	54.78	66.97	62.68
	W6A6	AWQ	59.83	57.78	61.48	58.94	52.95	64.25	59.83
		QLoRA	62.98	57.78	61.85	61.87	53.99	65.37	61.60
		Q-VLM	64.14	58.68	62.85	62.80	55.23	66.69	62.46
	W4A4	AWQ	53.69	49.58	54.27	52.52	47.81	57.65	52.98
		QLoRA	54.48	49.69	55.55	53.20	47.30	57.58	53.24
		Q-VLM	55.06	51.94	56.27	54.57	48.03	58.34	54.72

Table 2: Comparisons with the state-of-the-arts post-training quantization methods for LLaVA-v1.3 and MoE-LLaVA models across bitwidth setting. Results (accuracy) on Science QA dataset. Question classes: NAT = natural science, SOC = social science, LAN = language science, TXT = text context, IMG = image context, NO = no context.

Model	Dataset	FP	W6A6			W4A4		
			AWQ	QLoRA	Q-VLM	AWQ	QLoRA	Q-VLM
LLaVA-7B	SQA	66.79	65.87	66.16	66.67	56.73	56.50	57.70
	VizWiz	49.87	48.51	49.23	49.86	44.29	44.73	45.82
	VQA v2	78.50	77.51	77.58	78.52	71.89	72.02	72.51
LLaVA-13B	SQA	71.61	71.37	71.52	72.27	68.13	68.04	68.84
	VizWiz	53.63	52.35	52.85	53.69	47.55	47.97	49.28
	VQA v2	79.94	78.83	79.25	79.65	71.55	72.19	73.02

Table 3: Comparisons with the state-of-the-arts post-training quantization methods for LLaVA-v1.5 models in various VQA datasets across bitwidth setting.

Method	FP			W8A8			W4A4		
	Time	Memory	Accuracy	Time	Memory	Accuracy	Time	Memory	Accuracy
QLoRA				16.7h	16.5G	89.32	17.0h	10.7G	78.64
AWQ	12.9h	24.0G	90.00	11.2h	17.2G	88.94	8.9h	11.2G	77.91
Q-VLM				11.2h	15.7G	89.82	8.9h	9.6G	80.78

Table 4: Comparisons with the state-of-the-arts post-training quantization methods for LLaVA-v1.3-13B models about inference time, memory and accuracy in Science QA dataset.